

How to Select AA-Size Battery Cells

Electronic devices run on batteries of different shapes and sizes. But the most popular amongst them are AA-size primary cylindrical cells. These cells' prices are different but for most buyers they all are the same. But does a cell with higher price last longer? To find out we did a simple accelerated test

VIVEK PANCHABHAIYA

Modern battery cells use a variety of chemicals to power their reactions. Common battery chemistries include:

Zinc-carbon. These are the lowest-cost primary (non-rechargeable) cells meant for household use. The anode is zinc, the cathode is manganese dioxide and the electrolyte is ammonium chloride or zinc chloride. These cells produce very low power, but have a good shelf life and are well suited for clocks and remote controls.

Alkaline. These are the most commonly used primary cells. The cathode is made of manganese dioxide, while the anode is zinc powder. These cells have derived their name from the potassium hydroxide electrolyte, which is an alkaline substance. These provide more power-per-use than zinc-carbon and secondary (rechargeable) battery cells, and have an excellent shelf life.

Lithium. These are primary cells with a lithium metal or a lithium compound as an anode. These offer performance advantages well beyond the capabilities of conventional aqueous electrolyte battery systems. Their shelf life can be well above 10 years and they work at very low temperatures. These are mainly used in small formats (coin cells up to AA size) because bigger sizes of lithium batteries are a safety concern in consumer applications and are only used in military applications.

Silver oxide. These are also primary batteries with relatively very high energy/weight ratio. Their cost is linked to the price of silver. These are avail-



Fig. 1: Different sizes of batteries

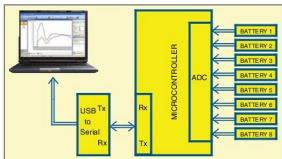


Fig. 2: Block diagram of the test setup

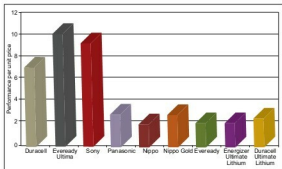


Fig. 3: Performance per unit price

able in either very small sizes as button cells, where the amount of silver used is small and not a significant contributor to the overall product cost, or in large custom-design batteries, where the superior performance characteristics of the silver oxide chemistry outweigh cost considerations.

Nickel-cadmium. These are rechargeable or secondary batteries

which use nickel oxide hydroxide and metallic cadmium as electrodes. The abbreviation Ni-Cd is derived from the chemical symbols of nickel (Ni) and cadmium (Cd). Ni-Cd batteries are rugged and reliable. These exhibit high-power capability, wide operating temperature range and long lifecycle, but have low runtime per charge.

Nickel-metal hydride. A nickel-metal hydride battery cell, abbreviated as NiMH or Ni-MH, is also a rechargeable cell. It uses positive electrodes of nickel oxyhydroxide (NiOOH), such as the NiCd, but the negative electrodes use a

TABLE I
Types of AA-Size Battery Cells

Types	IEC name	Nominal voltage	Rechargeable
Zinc-carbon	R6	1.5V	No
Alkaline	LR6	1.5V	No
Li-FeS ₂	FR6	1.5V	No
NiCd	KR6	1.2V	Yes
NiMH	HR6	1.2V	Yes
NiZn	ZR6	1.65V	Yes